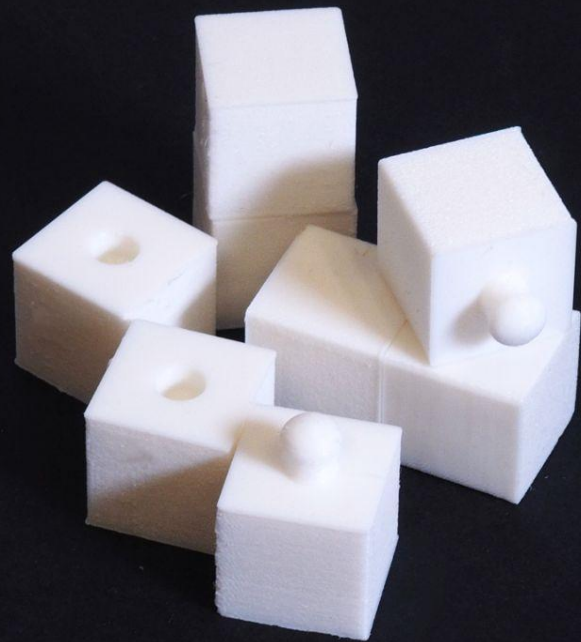




MOGO: Modular Soft Robotic Systems and Connection Techniques for Reconfigurable Assemblies

CMU 24-673 Final Presentation

Yujia Liu, Rohan Singh, Hongfei Liu, Brady Stump

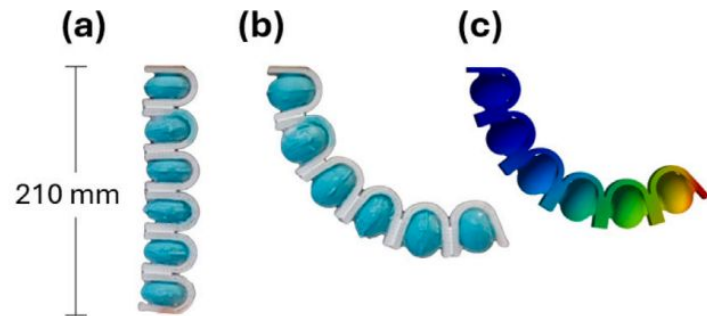
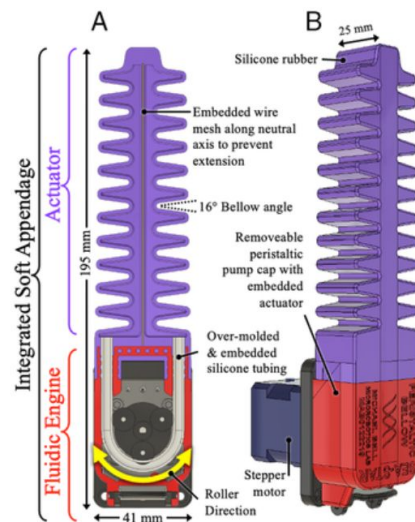
Short Recap

The Problem:

- Traditional robots are rigid, expensive, and unsafe around humans, soft robotics offers safe human interaction and adaptability
- But soft robots are difficult to design and fabricate
- Each application requires custom engineering from scratch

Our Vision:

- What if building soft robots was as intuitive as LEGOs?
- Enable rapid prototyping through modular, reconfigurable building blocks
- Make soft robotics accessible to researchers, educators, and makers



Our Journey

Where We Started:

Explored cable-driven actuation systems -
Investigated cable routing through flexible
modules

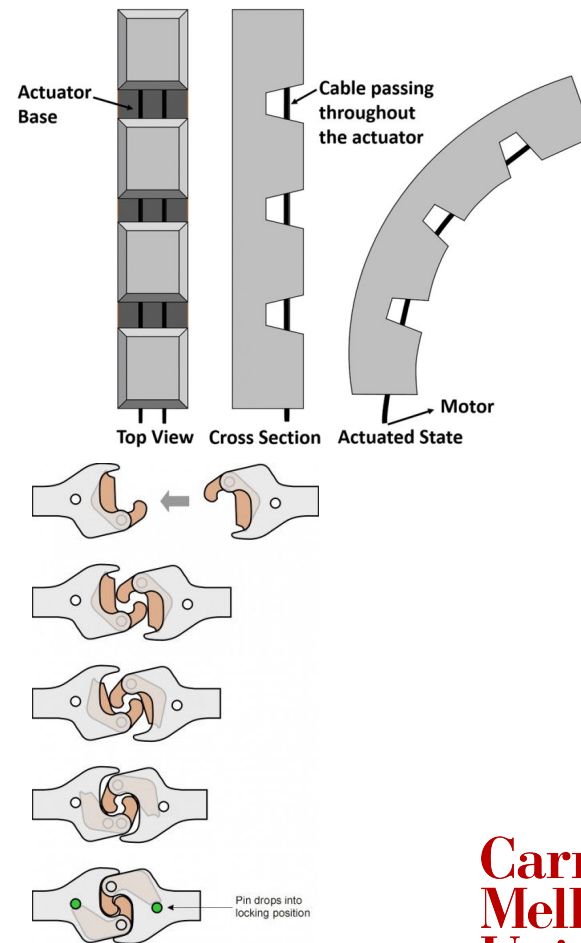
Focused on: "How do we make the actuation
work?"

The Pivot:

- Realized actuation wasn't the bottleneck
- The real challenge: How do modules connect?

Key Question:

"What makes LEGOs successful?" → Easy +
Robust connections that anyone can use



Manufacturing Phase 1

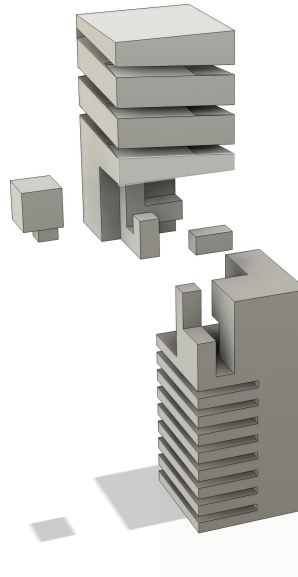
Equipments and Materials:

- Formlabs SLA 3D Printer & cleaning station
 - Formlabs Flexible Resin 80A
 - Isopropyl alcohol (IPA) for post-processing
 - UV curing light

Purpose:

Try interlocking structure

Diagrams:



Experiment:

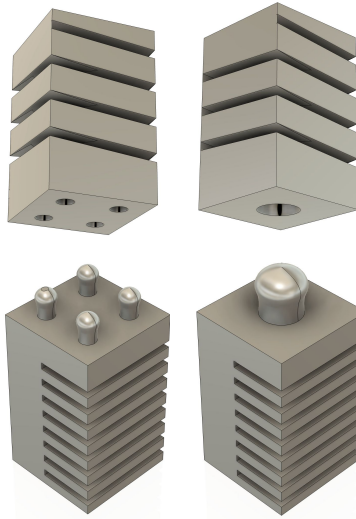


Manufacturing Phase 2

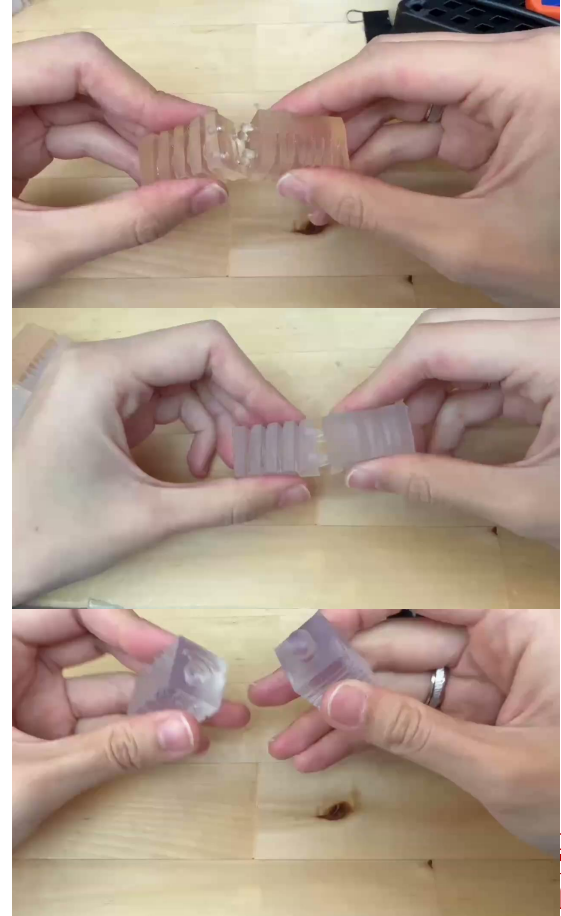
Equipments and Materials:

- Formlabs SLA 3D Printer & cleaning station
 - Formlabs Flexible Resin 80A
 - Isopropyl alcohol (IPA) for post-processing
 - UV curing light

Diagrams:



Experiment:



Purpose:

Simplify the structure

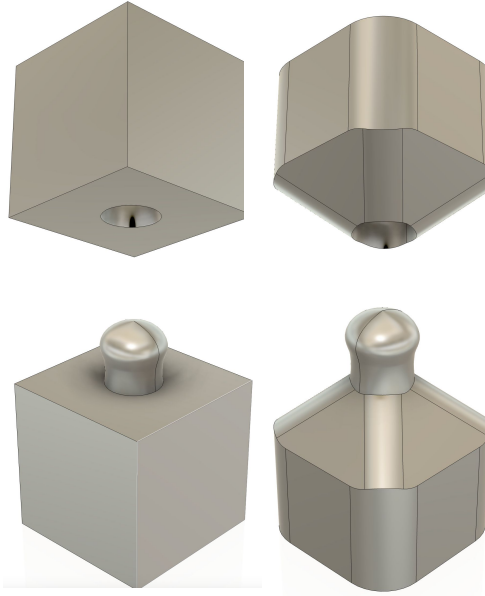
Try different parameters

Manufacturing Phase 3

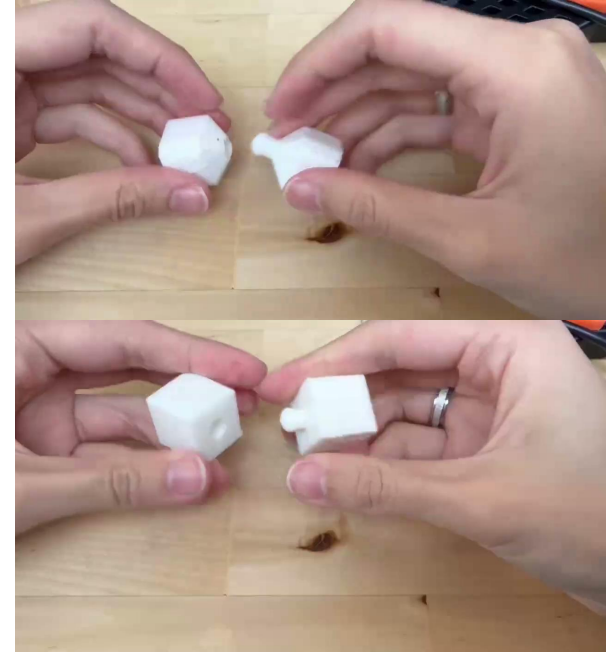
Equipments and Materials:

- Anycubic Kobra 3
 - TPU (Thermoplastic Polyurethane) 30% infill

Diagrams:



Experiment:

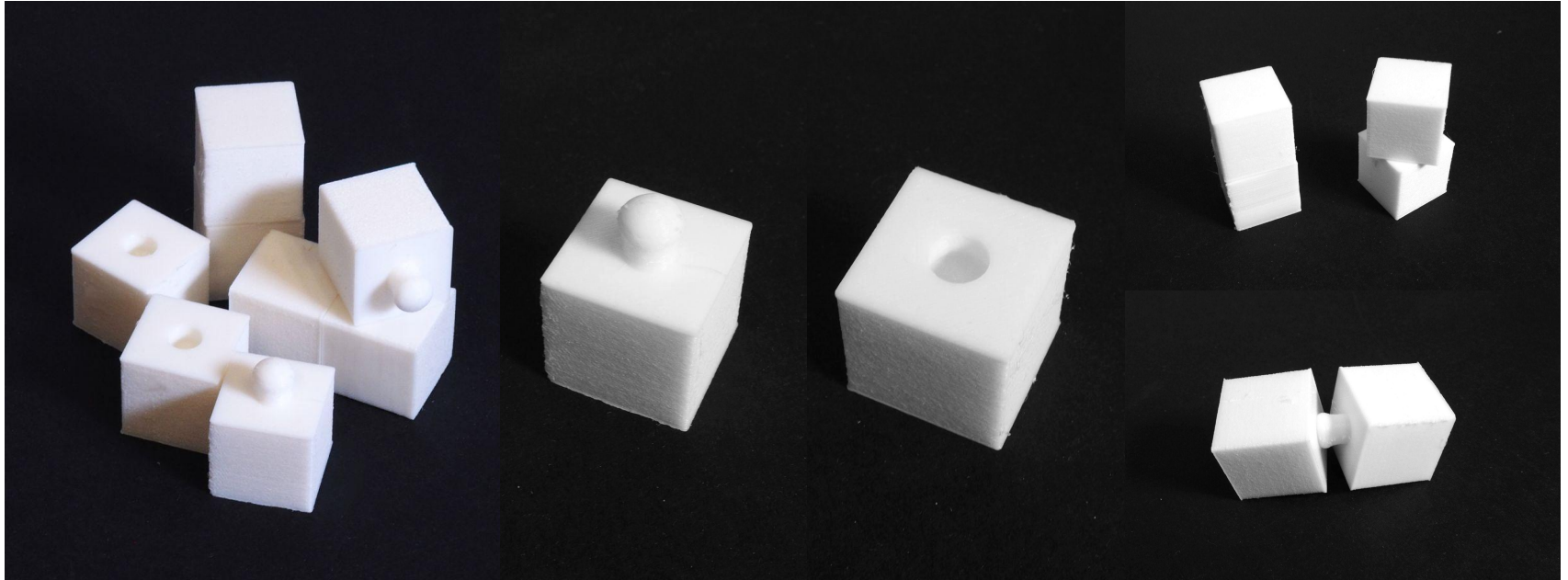


Purpose:

Simplify to focus only on the connection

Try different infill densities

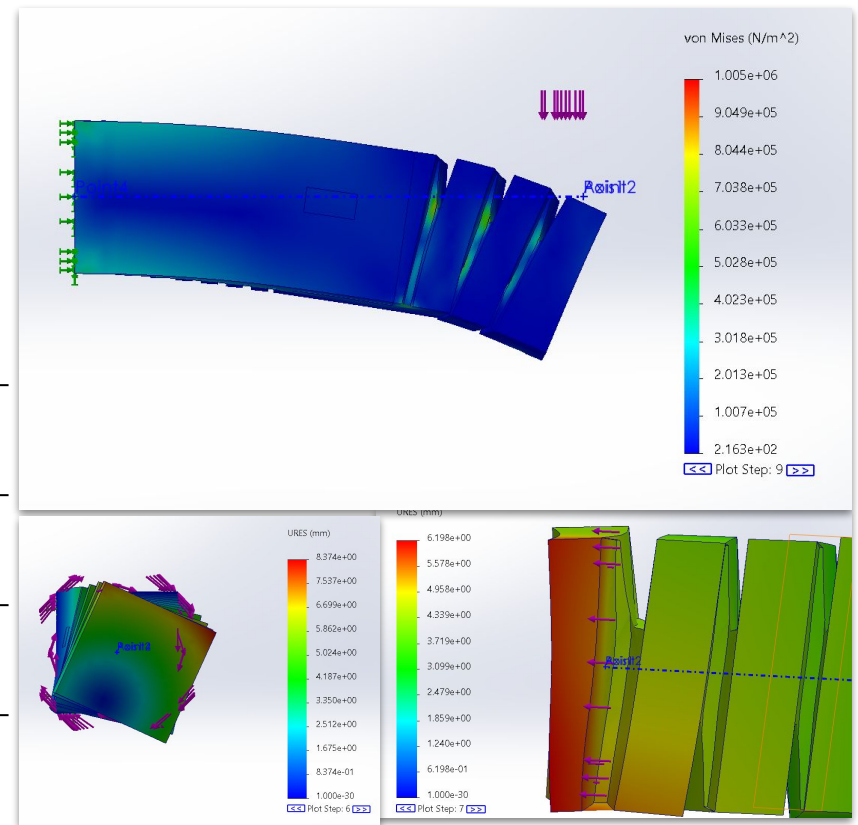
Manufacturing - Final Version



Negative Results

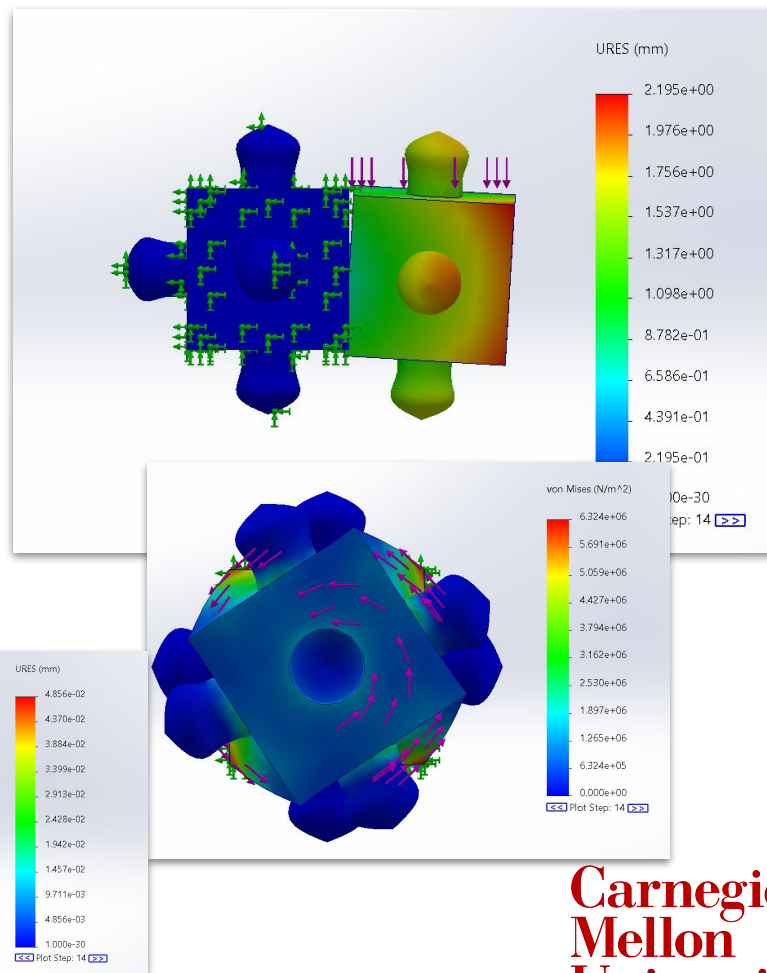
Very flexible but difficult to attach and not simplistic design

Failure	What Happened	Lesson Learned
Torsional weakness	Unlocks at 45° rotation	Need rotational constraint → add magnets
Manufacturing variance	±0.3mm in ribs, 15% failed first-try assemblies	Design for larger tolerances, QC checks
Combined loading	60% strength loss vs single-axis testing	Always test multi-modal stress states
Assembly force	12N peak (exceeded 10N target)	Balance strength vs. accessibility



Negative Results

Tried	Connector joints fabricated in TPU based on initial CAD specs (only tried 30% infill)
Result	Parts didn't fit; physical behavior diverged from simulation
Root Cause	Insufficient design tolerances + TPU hyperelasticity not modeled accurately
Lesson	Soft materials need 2-3× the tolerance of rigid parts; validate simulations with material testing first



Research Question Formation

Initial research question vs. Refined question.

- How can we design soft, modular, reconfigurable connectors that are easy to assemble and provide strong mechanical performance?
- Can a single-piece ribbed interference-fit connector maintain high mechanical strength across multi-axis loading while remaining user-assemblable?

What made the question testable/measurable?

- Assembly force → measured with a force gauge
- Tensile, torsional, and bending strength → measured on the mechanical testing platform?
- User experience → quantified by assembly failure rate and user error modes

What could be improved?

- Early weeks were spent exploring overly complicated multi-part designs.
- Too much time was invested in designs that were hardly printable or testable.
- Mechanical testing ramped up late—the project would benefit from running preliminary tests earlier.

Key Takeaways

Insight #1: Complex multi-part ideas should be abandoned quickly in the beginning

Insight #2: Single-parameter iteration is essential for interpretable results

Insight #3: Non-essential features must be cut early during process going

Individual Contributions

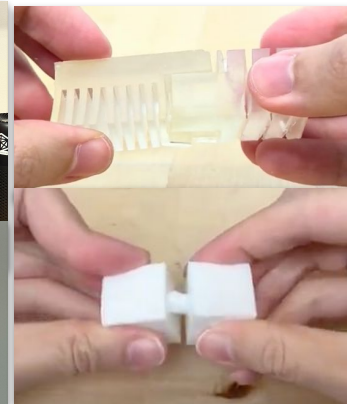
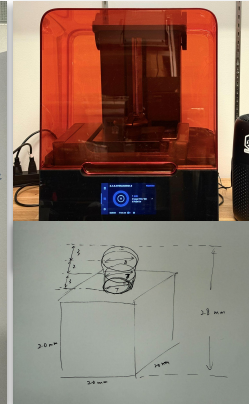
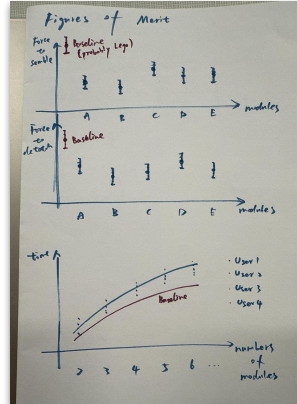
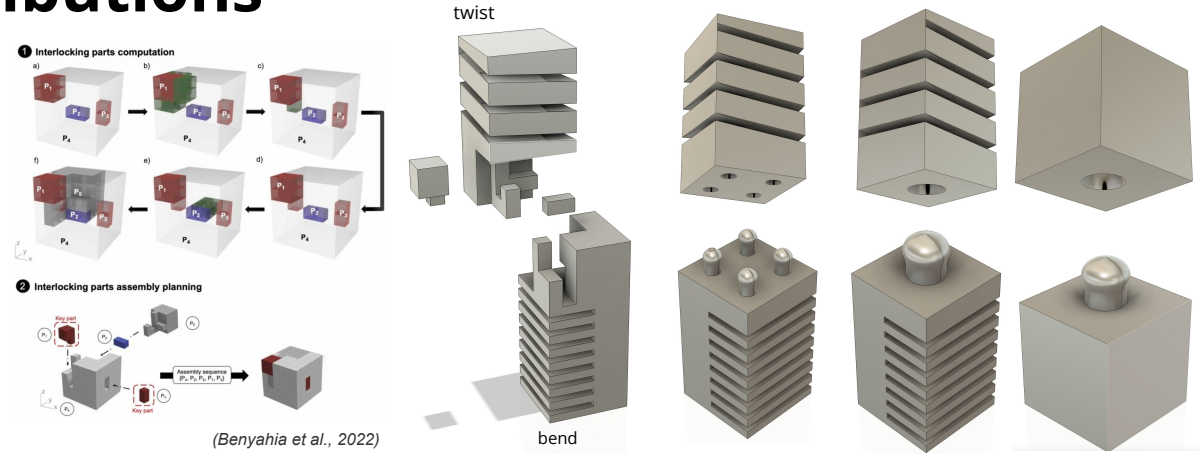
Yujia

Ideation

Writing

3D modeling & 3D Printing

Manufacturing Process



Individual Contributions

Rohan

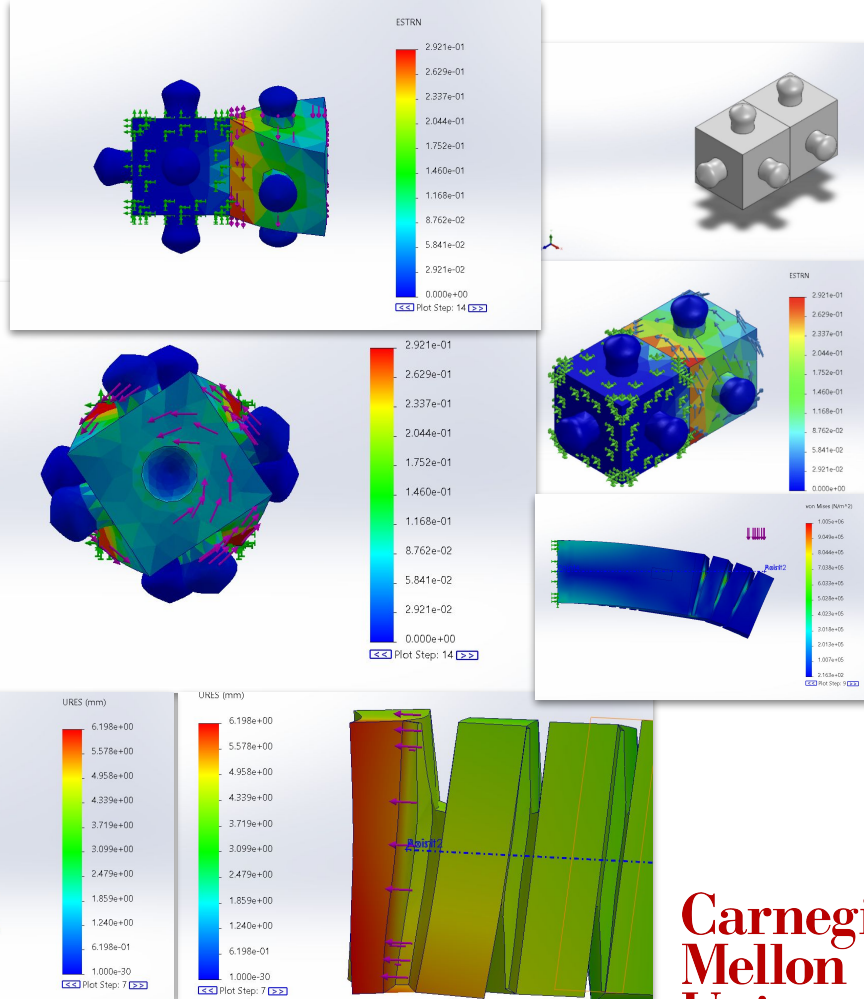
Literature researching

Ideation

FEA simulation work

Midterm Report

Investigation



Individual Contributions

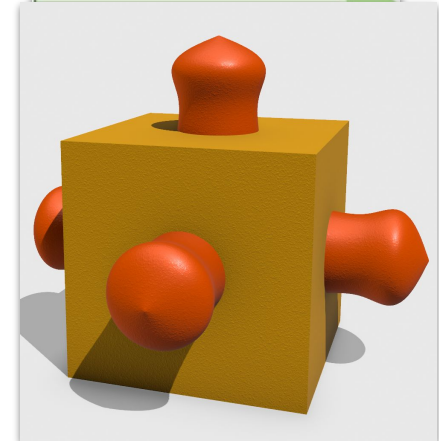
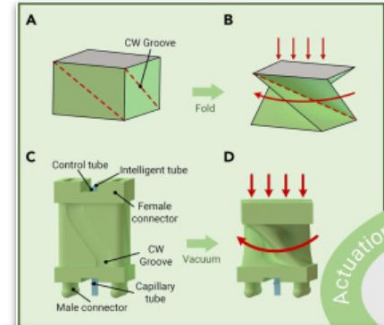
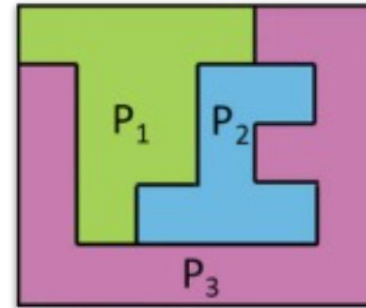
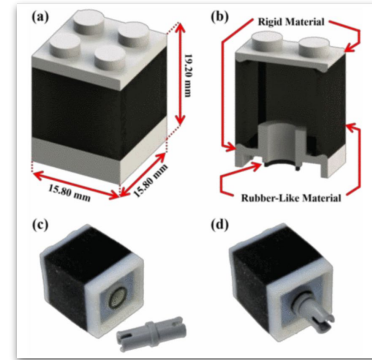
Hongfei

Literature researching

Ideation & Conceptualization

3D modeling

Writing



**Carnegie
Mellon
University**

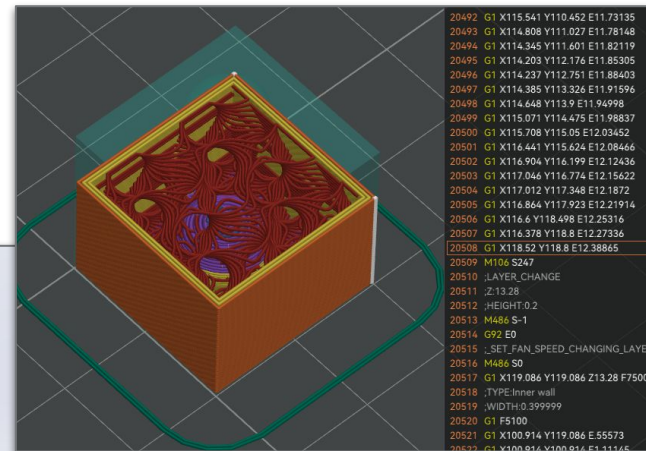
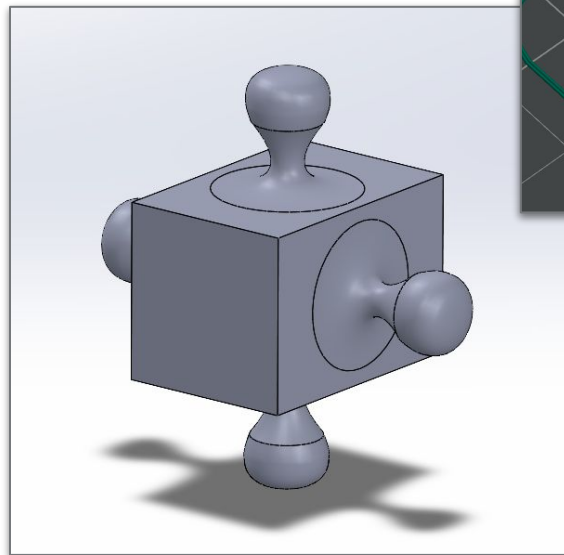
Individual Contributions

Brady

Mechanical Ideation

FDM Expertise

3D Modeling



The logo for Carnegie Mellon University is positioned on the left side of the slide. It features the university's name in a white, serif font, stacked in three lines: "Carnegie", "Mellon", and "University". The text is set against a dark blue background that is decorated with a complex pattern of intersecting diagonal lines in various colors, including red, green, and yellow, creating a grid-like effect.

**Carnegie
Mellon
University**

Thank you!
Questions?

Citations

1. Zhang, H., et al. (2020). "Modular Soft Robotics: Modular Units, Connection Mechanisms, and Applications." *Advanced Intelligent Systems*, 2(1), 1900166.
2. Science Advances (2024). "Pneumatic coding blocks enable programmability of electronics-free fluidic soft robots." *Science Advances*, doi: 10.1126/sciadv.adr2433
3. Nature Communications (2023). "Embedded shape morphing for morphologically adaptive robots." *Nature Communications*, 14, Article number: 6843.
4. Communications Engineering (2022). "Self-replicating hierarchical modular robotic swarms." *Nature Communications Engineering*, 1, Article number: 34.
5. ScienceDirect (2024). "Modelling of Modular Soft Robots: From a Single to Multiple Building Blocks." *Robotics and Autonomous Systems*.
6. Nature Materials (2025). "Scalable functionalized shape memory alloy fiber with synergistic effect for robotic hand and microrobot." *npj Flexible Electronics*.
7. Science Robotics (2022). "A robotic honeycomb for interaction with a honeybee colony." *Science Robotics*, 7(66), eadd7385.
8. Nature Communications (2024). "Bio-inspired autonomy in soft robots." *Communications Materials*, 5, Article number: 287.
9. Advanced Functional Materials (2014). "Pneumatic Networks for Soft Robotics that Actuate Rapidly." *Advanced Functional Materials*, 24(15), 2163-2170.
10. Frontiers in Robotics and AI (2021). "Hardware Methods for Onboard Control of Fluidically Actuated Soft Robots." *Frontiers in Robotics and AI*, 8, 720702.
11. Smart Materials and Structures (2024). "Reconfigurable modular soft robots with modulating stiffness and versatile task capabilities." *Smart Materials and Structures*, 33(6), 065040. doi: 10.1088/1361-665X/ad4d35.